

Chapter 16

Continuity and Differentiability

Only One Option Correct Type Questions

1. Let $f(x)$ be differentiable on the interval $(0, \infty)$ such that $f(1) = 1$, and $\lim_{t \rightarrow x} \frac{t^2 f(x) - x^2 f(t)}{t - x} = 1$ for each $x > 0$. Then $f(x)$ is [IIT-JEE-2007 (Paper-1)]
- (A) $\frac{1}{3x} + \frac{2x^2}{3}$ (B) $\frac{-1}{3x} + \frac{4x^2}{3}$
(C) $\frac{-1}{x} + \frac{2}{x^2}$ (D) $\frac{1}{x}$
2. Let $g(x) = \frac{(x-1)^n}{\log \cos^m(x-1)}$; $0 < x < 2$, m and n are integers, $m \neq 0$, $n > 0$, and let p be the left hand derivative of $|x-1|$ at $x=1$. If $\lim_{x \rightarrow 1^+} g(x) = p$, then [IIT-JEE-2008 (Paper-1)]
- (A) $n = 1, m = 1$ (B) $n = 1, m = -1$
(C) $n = 2, m = 2$ (D) $n > 2, m = n$
3. Let $f(x) = \begin{cases} x^2 \left| \cos \frac{\pi}{x} \right|, & x \neq 0 \\ 0, & x = 0 \end{cases}$, $x \in \mathbb{R}$ then f is [IIT-JEE-2012 (Paper-1)]
- (A) Differentiable both at $x = 0$ and at $x = 2$
(B) Differentiable at $x = 0$ but not differentiable at $x = 2$
(C) Not differentiable at $x = 0$ but differentiable at $x = 2$
(D) Differentiable neither at $x = 0$ nor at $x = 2$

One or More Option(s) Correct Type Questions

4. Let f be a real-valued function defined on the interval $(0, \infty)$ by $f(x) = \ln x + \int_0^x \sqrt{1 + \sin t} dt$. Then which of the following statement(s) is/are true? [IIT-JEE-2010 (Paper-1)]
- (A) $f''(x)$ exists for all $x \in (0, \infty)$
(B) $f'(x)$ exists for all $x \in (0, \infty)$ and f' is continuous on $(0, \infty)$, but not differentiable on $(0, \infty)$
(C) There exists $\alpha > 1$ such that $|f'(x)| < |f(x)|$ for all $x \in (\alpha, \infty)$
(D) There exists $\beta > 0$ such that $|f(x)| + |f'(x)| \leq \beta$ for all $x \in (0, \infty)$

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x + y) = f(x) + f(y)$, $\forall x, y \in \mathbb{R}$. If $f(x)$ is differentiable at $x = 0$, then [IIT-JEE-2011 (Paper-1)]
- (A) $f(x)$ is differentiable only in a finite interval containing zero
- (B) $f(x)$ is continuous $\forall x \in \mathbb{R}$
- (C) $f'(x)$ is constant $\forall x \in \mathbb{R}$
- (D) $f(x)$ is differentiable except at finitely many points

6. If $f(x) = \begin{cases} -x - \frac{\pi}{2}, & x \leq -\frac{\pi}{2}, \\ -\cos x, & -\frac{\pi}{2} < x \leq 0, \\ x - 1, & 0 < x \leq 1, \\ \ln x, & x > 1, \end{cases}$ then [IIT-JEE-2011 (Paper-2)]

- (A) $f(x)$ is continuous at $x = -\frac{\pi}{2}$
- (B) $f(x)$ is not differentiable at $x = 0$
- (C) $f(x)$ is differentiable at $x = 1$
- (D) $f(x)$ is differentiable at $x = -\frac{3}{2}$

7. For every integer n , let a_n and b_n be real numbers. Let function $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = \begin{cases} a_n + \sin \pi x, & \text{for } x \in [2n, 2n+1] \\ b_n + \cos \pi x, & \text{for } x \in (2n-1, 2n) \end{cases},$$

for all integers n . If f is continuous, then which of the following hold(s) for all n ? [IIT-JEE-2012 (Paper-2)]

- (A) $a_{n-1} - b_{n-1} = 0$
- (B) $a_n - b_n = 1$
- (C) $a_n - b_{n+1} = 1$
- (D) $a_{n-1} - b_n = -1$

8. Let $f : [a, b] \rightarrow [1, \infty)$ be a continuous function and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$g(x) = \begin{cases} 0 & \text{if } x < a \\ \int_a^x f(t) dt & \text{if } a \leq x \leq b \\ \int_a^b f(t) dt & \text{if } x > b \end{cases}.$$
 Then

[JEE (Adv)-2014 (Paper-1)]

- (A) $g(x)$ is continuous but not differentiable at a
- (B) $g(x)$ is differentiable on $\mathbb{R}$
- (C) $g(x)$ is continuous but not differentiable at b
- (D) $g(x)$ is continuous and differentiable at either a or b but not both

9. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $g(0) = 0$, $g'(0) = 0$ and $g'(1) \neq 0$. Let

$$f(x) = \begin{cases} \frac{x}{|x|} g(x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

and $h(x) = e^{|x|}$ for all $x \in \mathbb{R}$. Let $(f \circ h)(x)$ denote $f(h(x))$ and $(h \circ f)(x)$ denote $h(f(x))$. Then which of the following is (are) true? [JEE (Adv)-2015 (Paper-1)]

- (A) f is differentiable at $x = 0$
- (B) h is differentiable at $x = 0$
- (C) $f \circ h$ is differentiable at $x = 0$
- (D) $h \circ f$ is differentiable at $x = 0$

10. Let $a, b \in \mathbb{R}$ and $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = a \cos(|x^3 - x|) + b|x| \sin(|x^3 + x|)$. Then f is

[JEE (Adv)-2016 (Paper-2)]

- (A) Differentiable at $x = 0$ if $a = 0$ and $b = 1$
 (B) Differentiable at $x = 1$ if $a = 1$ and $b = 0$
 (C) **NOT** differentiable at $x = 0$ if $a = 1$ and $b = 0$
 (D) **NOT** differentiable at $x = 1$ if $a = 1$ and $b = 1$

11. Let $f: \left[-\frac{1}{2}, 2\right] \rightarrow \mathbb{R}$ and $g: \left[-\frac{1}{2}, 2\right] \rightarrow \mathbb{R}$ be functions defined by $f(x) = [x^2 - 3]$ and $g(x) = |x|f(x) + |4x - 7|f(x)$, where $[y]$ denotes the greatest integer less than or equal to y for $y \in \mathbb{R}$. Then

[JEE (Adv)-2016 (Paper-2)]

- (A) f is discontinuous exactly at three points in $\left[-\frac{1}{2}, 2\right]$
 (B) f is discontinuous exactly at four points in $\left[-\frac{1}{2}, 2\right]$
 (C) g is **NOT** differentiable exactly at four points in $\left(-\frac{1}{2}, 2\right)$
 (D) g is **NOT** differentiable exactly at five points in $\left(-\frac{1}{2}, 2\right)$

12. Let $f: \mathbb{R} \rightarrow (0, 1)$ be a continuous function. Then, which of the following function(s) has(have) the value zero at some point in the interval $(0, 1)$?

[JEE (Adv)-2017 (Paper-1)]

- (A) $f(x) + \int_0^{\frac{\pi}{2}} f(t) \sin t \, dt$ (B) $x^9 - f(x)$
 (C) $x - \int_0^{\frac{\pi}{2-x}} f(t) \cos t \, dt$ (D) $e^x - \int_0^x f(t) \sin t \, dt$

13. Let $[x]$ be the greatest integer less than or equals to x . Then, at which of the following point(s) the function $f(x) = x \cos(\pi(x + [x]))$ is discontinuous?

[JEE (Adv)-2017 (Paper-1)]

- (A) $x = 0$ (B) $x = 1$
 (C) $x = 2$ (D) $x = -1$

14. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be two non-constant differentiable functions. If $f'(x) = (e^{(f(x)-g(x))}) g'(x)$ for all $x \in \mathbb{R}$, and $f(1) = g(2) = 1$, then which of the following statement(s) is (are) TRUE?

[JEE (Adv)-2018 (Paper-1)]

- (A) $f(2) < 1 - \log_e 2$ (B) $f(2) > 1 - \log_e 2$
 (C) $g(1) > 1 - \log_e 2$ (D) $g(1) < 1 - \log_e 2$

15. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function. We say that f has

PROPERTY 1 if $\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{\sqrt{|h|}}$ exists and is finite, and

PROPERTY 2 if $\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h^2}$ exists and is finite.

Then which of the following options is/are correct?

[JEE (Adv)-2019 (Paper-2)]

(A) $f(x) = x^{2/3}$ has PROPERTY 1

(B) $f(x) = |x|$ has PROPERTY 1

(C) $f(x) = x|x|$ has PROPERTY 2

(D) $f(x) = \sin x$ has PROPERTY 2

16. Let the function $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^3 - x^2 + (x - 1)\sin x$ and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be an arbitrary function. Let $fg : \mathbb{R} \rightarrow \mathbb{R}$ be the product function defined by $(fg)(x) = f(x)g(x)$. Then which of the following statements is/are TRUE?

[JEE (Adv)-2020 (Paper-1)]

(A) If g is continuous at $x = 1$, then fg is differentiable at $x = 1$

(B) If fg is differentiable at $x = 1$, then g is continuous at $x = 1$

(C) If g is differentiable at $x = 1$, then fg is differentiable at $x = 1$

(D) If fg is differentiable at $x = 1$, then g is differentiable at $x = 1$

17. Let $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow \mathbb{R}$ be a continuous function such that $f(0) = 1$ and $\int_0^{\frac{\pi}{3}} f(t) dt = 0$

Then which of the following statements is (are) TRUE?

[JEE (Adv)-2021 (Paper-2)]

(A) The equation $f(x) - 3 \cos 3x = 0$ has at least one solution in $\left(0, \frac{\pi}{3}\right)$

(B) The equation $f(x) - 3 \sin 3x = -\frac{6}{\pi}$ has at least one solution in $\left(0, \frac{\pi}{3}\right)$

(C) $\lim_{x \rightarrow 0} \frac{x \int_0^x f(t) dt}{1 - e^{x^2}} = -1$

(D) $\lim_{x \rightarrow 0} \frac{\sin x \int_0^x f(t) dt}{x^2} = -1$

Matrix-Match / Matrix Based Type Questions

18. Let $f_1 : \mathbb{R} \rightarrow \mathbb{R}$, $f_2 : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$, $f_3 : \left(-1, e^{\frac{\pi}{2}} - 2\right) \rightarrow \mathbb{R}$ and $f_4 : \mathbb{R} \rightarrow \mathbb{R}$ be functions defined by

[JEE (Adv)-2018 (Paper-2)]

(i) $f_1(x) = \sin\left(\sqrt{1 - e^{-x^2}}\right)$,

(ii) $f_2(x) = \begin{cases} \frac{|\sin x|}{\tan^{-1} x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$, where the inverse trigonometric function $\tan^{-1} x$ assumes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(iii) $f_3(x) = [\sin(\log_e(x + 2))]$, where, for $t \in \mathbb{R}$, $[t]$ denotes the greatest integer less than or equal to t ,

(iv) $f_4(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

List-IP. The function f_1 isQ. The function f_2 isR. The function f_3 isS. The function f_4 is

The correct option is

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$ (C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$ **List-II**1. NOT continuous at $x = 0$ 2. continuous at $x = 0$ and NOT differentiable at $x = 0$ 3. differentiable at $x = 0$ and its derivative is NOT continuous at $x = 0$ 4. differentiable at $x = 0$ and its derivative is continuous at $x = 0$ (B) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$ (D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$ **Integer / Numerical Value Type Questions**19. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Define $h : \mathbb{R} \rightarrow \mathbb{R}$ by

$$h(x) = \begin{cases} \max \{f(x), g(x)\} & \text{if } x \leq 0, \\ \min \{f(x), g(x)\} & \text{if } x > 0. \end{cases}$$

The number of points at which $h(x)$ is not differentiable is**[JEE (Adv)-2014 (Paper-1)]**20. Let e denote the base of the natural logarithm. The value of the real number a for which the right hand limit

$$\lim_{x \rightarrow 0^+} \frac{(1-x)^{\frac{1}{x}} - e^{-1}}{x^a}$$

is equal to a nonzero real number, is _____

[JEE (Adv)-2020 (Paper-1)]21. Let the functions $f : (-1, 1) \rightarrow \mathbb{R}$ and $g : (-1, 1) \rightarrow (-1, 1)$ be defined by

$$f(x) = |2x - 1| + |2x + 1| \text{ and } g(x) = x - [x],$$

where $[x]$ denotes the greatest integer less than or equal to x . Let $f \circ g : (-1, 1) \rightarrow \mathbb{R}$ be the composite function defined by $(f \circ g)(x) = f(g(x))$. Suppose c is the number of points in the interval $(-1, 1)$ at which $f \circ g$ is NOT continuous, and suppose d is the number of points in the interval $(-1, 1)$ at which $f \circ g$ is NOT differentiable. Then the value of $c + d$ is _____

[JEE (Adv)-2020 (Paper-2)]22. Let α be a positive real number. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : (\alpha, \infty) \rightarrow \mathbb{R}$ be the functions defined by

$$f(x) = \sin\left(\frac{\pi x}{12}\right) \text{ and } g(x) = \frac{2 \log_e(\sqrt{x} - \sqrt{\alpha})}{\log_e(e^{\sqrt{x}} - e^{\sqrt{\alpha}})}.$$

Then the value of $\lim_{x \rightarrow \alpha^+} f(g(x))$ is _____.**[JEE (Adv)-2022 (Paper-1)]**